P99. 题1.

0.5s 的调节时间

$$T_s = \frac{4}{\xi \omega_n} = 0.5$$

15%超调量

$$M_p = e^{-\xi\pi/\sqrt{1-\xi^2}} = 15\%$$

$$\frac{-\xi\pi}{\sqrt{1-\xi^2}} = \ln 0.15$$

$$\xi^2\pi^2 = (\ln 0.15)^2 \cdot (1-\xi^2)$$

$$\xi^2\left[\pi^2+(\ln 0.15)^2\right] = (\ln 0.15)^2$$

$$\xi = \frac{-\ln 0.15}{\sqrt{\pi^2+(\ln 0.15)^2}}$$

$$= 0.517$$

$\omega_n = 15.474$

⑰

$$K \cdot \frac{w_n^2}{s^2 + 2\zeta w_n s + w_n^2}$$

$$= \frac{15.474^2 \cdot K}{s^2 + 2 \times 0.517 \times 15.474 s + 15.474^2}$$

$$= \frac{239.44 K}{s^2 + 16 s + 239.44}$$

$$s^2 + 16 s + 239.44 = 0$$

$$s_1 = -8 + 13.245 j$$

$$s_2 = -8 - 13.245 j$$

$$\frac{239.44}{[s - (-8 + 13.245 j)][s - (-8 - 13.245 j)]}$$

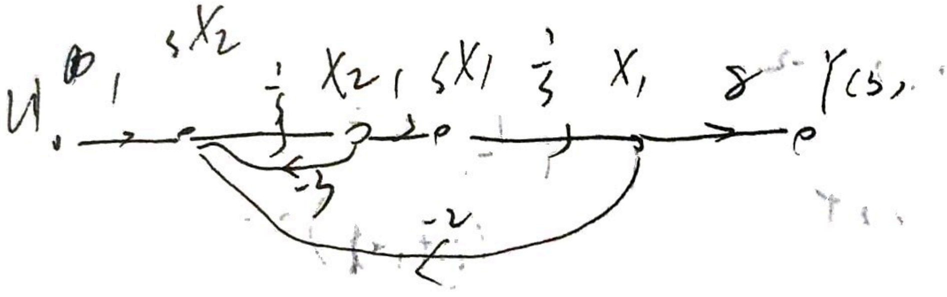
$$G(s) = \frac{8}{(s+1)(s+2)} = \frac{8}{s^2+3s+2}$$

求(2):

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \qquad \dot{x} = Ax + Bu$$

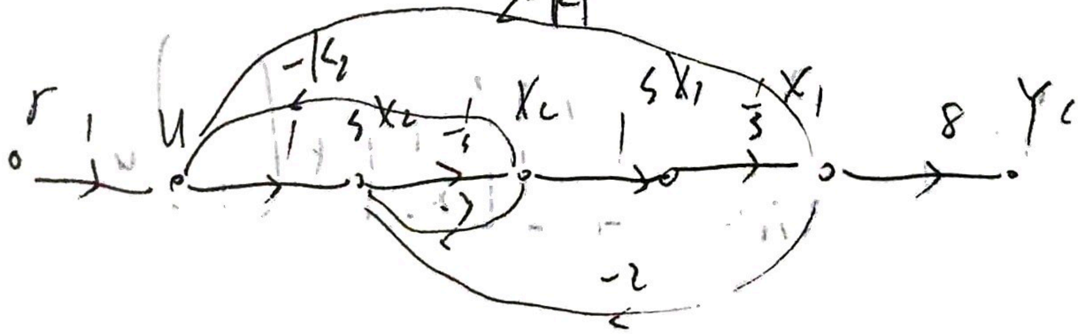
$$y = C^T x$$

$$y = [8 \;\; 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$



$$\dot{x} = Ax + B(r - Kx) \qquad u \to r - Kx$$

$$(A - BK)x + Br \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix} [k_1 \;\; k_2]$$



$$\begin{bmatrix} 0 & 1 \\ -(2+k_1) & -(3+k_2) \end{bmatrix}$$

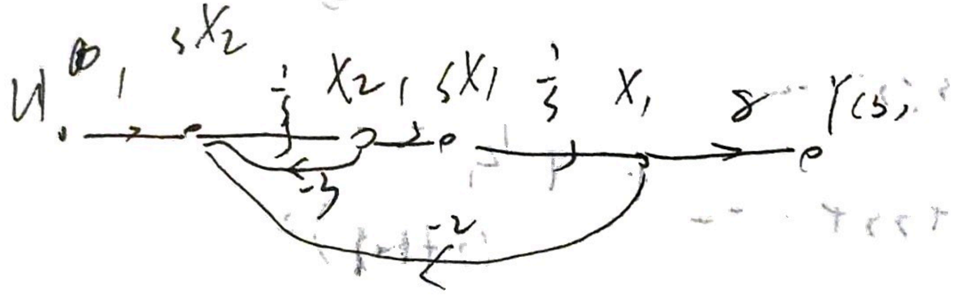
$$G(s) = \frac{8}{(s+1)(s+2)} = \frac{8}{s^2+3s+2}$$

求(2):

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ -2 & -3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ 1\end{bmatrix}u \qquad \dot{x} = Ax + Bu$$

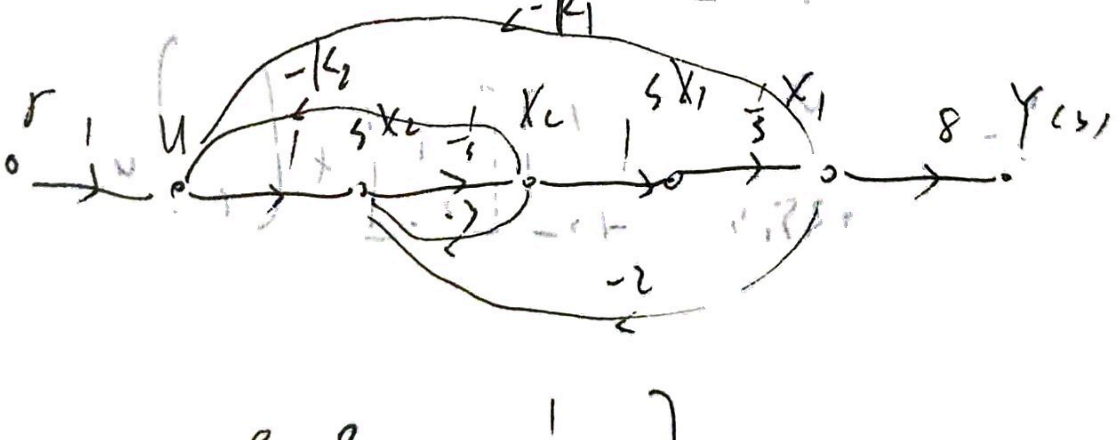
$$y = C^T x$$

$$y = [8, 0]\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$



$$\dot{x} = Ax + B(r - Kx) \qquad u \to r - Kx$$

$$(A-BK)x + Br \quad \begin{bmatrix}0\\ 1\end{bmatrix}[k_1, k_2]$$



$$\begin{bmatrix}0 & 1\\ -(2+k_1) & -(3+k_2)\end{bmatrix}$$

$\det[s] - 1s \cdot$ [illegible]

$$\begin{bmatrix} 0 & \\ -(2+k_1) & -(3+k_2) \end{bmatrix}$$

$$\begin{bmatrix} s & -1 \\ 2+k_1 & s+3+k_2 \end{bmatrix}$$

$s(s+3+k_2) - (-1)(2+k_1) = 0$

$s^2 + 3s + sk_2 + 2 + k_1 = 0$

$s^2 + (3+k_2)s + (2+k_1) = 0$

$3 + k_2 = 16$

$2 + k_1 = 239.44$

$\rightarrow \begin{cases} k_1 = 237.44 \\ k_2 = 13 \end{cases}$

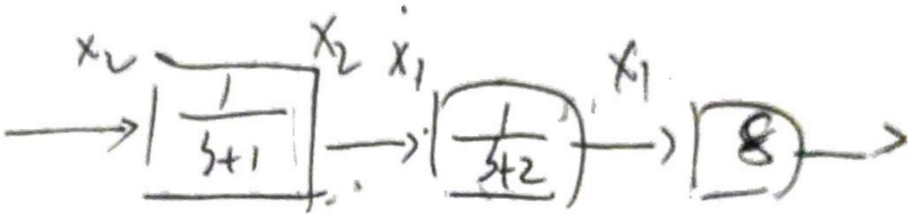
$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -235.44 & -16 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u$$

错了，这个搞定量型的.

100

$$\frac{\quad}{(s+1)(s+2)}$$

串联分解



$U(s)\cdot\frac{1}{s+1} = X_2(s)$

$U(s) = sX_2(s) + X_2(s)$

$sX_2(s) = -X_2(s) + U(s)$

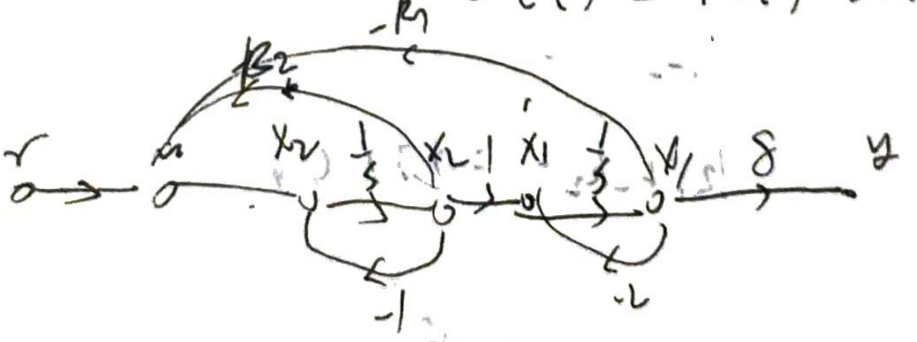
$= [0, -1]\begin{bmatrix}x_1(s)\\X_2(s)\end{bmatrix} + 1\,U(s)$

$R(s)\cdot\frac{1}{s+2} = C(s)$

$R(s) = (s+2)C(s)$

$r(t) = \dot{c}(t) + 2c(t)$

$\dot{c}(t) = r(t) - 2c(t)$



$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}-2 & 1\\ 0 & -1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ 1\end{bmatrix}u$$

$$A: \begin{pmatrix} -2 & 1 \\ -k_1 & -1-k_2 \end{pmatrix}$$

$sI - A$

$$\begin{pmatrix} s+2 & -1 \\ k_1 & s+1+k_2 \end{pmatrix}$$

$(s+2)(s+1+k_2)+k_1=0$

$s^2+2s+(1+k_2)s+k_1+2(1+k_2)=0$

$s^2+(3+k_2)s+(k_1+2k_2+2)=0$

$$\begin{cases} 3+k_2=16 \\ k_1+2k_2+2=239.44 \end{cases}$$

$$\begin{cases} k_1=211.5 \\ k_2=13 \end{cases}$$

p.99. 2. 设计全阶观测器.

算一下，我思考一下这个通路稳态输出多少.

如果输入 $u(t)=4$，阶跃

$u(t)=4$ $$\frac{8}{(s+1)(s+2)}$$

令：$\dot{x}=Ax+Bu$

$y=Cx$ $\quad s^2+3s+2$

用相变量型 $A\begin{bmatrix}0 & 1\\ -2 & -3\end{bmatrix}$ $B\begin{bmatrix}0\\ 1\end{bmatrix}$ $C[1,0]$ $D0$

稳态：$\dot{\vec{x}}=0$

$\dot{x}_1(t)=x_2(t)$

$\dot{x}_2(t)=-2x_1(t)-3x_2(t)+u(t)$

~~$x(t)=0$~~ $x_{2ss}=0$ $\quad -2x_{1ss}+4=0$

$x_{1ss}=2$

$y_{ss}=8x_1=16$

u(t) = 1(t) 阶跃

用 状态空间

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}-2 & 1\\ 0 & -1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ 1\end{bmatrix}u$$

$$y = \begin{bmatrix}8 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

无加反馈

$$\dot{x}_{1ss} = 0 = -2x_{1ss} + x_{2ss}$$

$$\dot{x}_{2ss} = 0 = 0\,x_{1ss} + -1\,x_{2ss} + u_{ss}$$

$$\rightarrow \begin{cases} x_{1ss} = 2 \\ x_{2ss} = 4 \end{cases}$$

$$y_{ss} = 8x_{1ss} = 8 \times 2 = 16$$

加反馈

$$\dot{x}_{1ss} = 0 = -2x_{1ss} + x_{2ss}$$

$$\dot{x}_{2ss} = 0 = -k_1 x_{1ss} + (-1-k_2)x_{2ss} + u_{ss}$$

x_{1ss} 用 k_1, k_2 表示

$$x_{2ss} = 2x_{1ss}$$

$$-k_1x_1 + (-1-k_2)\cdot 2x_1 + 4 = 0$$

$$x_{1ss} = \frac{4}{k_1+2k_2+2}$$

$$y_{ss} = 8 \times \frac{4}{k_1+2k_2+2}$$

$k_1 = 211.5$
$k_2 = 13$代

(104)

$y_{ss} = 0.1336$ ✓